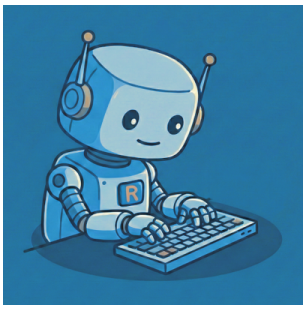
Nordheim Online



Rheolwyr

Installation and User Guide

Version 0.5.18 (with the October 2026 fixes) · **Platforms** Linux (GNOME, COSMIC; Wayland and X11) · **Updated** October 2026

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1 About Rheolwyr

Rheolwyr (Welsh for "manager", pronounced roughly *HRE-aw-loor*) is a text expander for Linux. You type a short trigger, such as `;sig`, and Rheolwyr replaces it with longer text, such as your email signature, in almost any application. It's built for GNOME and Pop!_OS COSMIC, on Wayland or X11.

Rheolwyr is the Linux counterpart of Shine, Nordheim Online's text expander for the Mac.

What Rheolwyr does

- Stores your snippets (a name, a trigger and the text) in a local database.
- Watches for triggers as you type and replaces them with your snippet.
- Restores your previous clipboard text after pasting long snippets.
- Backs up and restores snippets as JSON.

Rheolwyr is free, open-source software under the GNU General Public License v3. Source: github.com/TaliskerMan/Rheolwyr.

2 Read this first: what Rheolwyr needs to see

> **Important:** To spot your triggers, a text expander must read every key you press. On **Wayland**, Rheolwyr reads the keyboard directly from `/dev/input`, which requires adding your account to the **`input` group**. That membership lets your account read **every keystroke from every application, system-wide and permanently**, not only while Rheolwyr runs, until you remove yourself from the group.

What Rheolwyr does to limit the risk:

- It **never records or logs keystrokes** or snippet text. It keeps only the last 50 characters in memory to match triggers.
- It has **no network code**. Nothing is sent anywhere.
- The package **never adds you to the group automatically**. You choose to.

Least-privilege alternatives, for people who'd rather not make their login account able to read all input:

- **A dedicated service user** in the `input` and `uinput` groups that runs the listener, instead of your own account.
- **A udev rule for just your keyboard**, granting read access to that one device instead of all input devices.
- **An X11 session**, where Rheolwyr uses X11 input and doesn't need the `input` group (X11 has its own security trade-offs).

The repository's `SECURITY.md` explains these options in more detail.

3 System requirements

Operating system	Debian, Ubuntu, Pop!_OS or a derivative
Desktop	GNOME or COSMIC, Wayland or X11
Installed with the package	Python 3, GTK 4 and Libadwaita bindings, `python3-evdev`, `python3-pynput`, `wl-clipboard` or `xclip`
Wayland only	Your account in the `input` and `uinput` groups

4 Installing Rheolwyr

1. Download `rheolwyr_<version>_all.deb` from the release page.
2. Install it:

```
sudo apt install ./rheolwyr_0.5.18-1_all.deb
```

The installer creates a `uinput` group, installs a udev rule so that group can use the virtual keyboard (`/dev/uinput`), loads the `uinput` kernel module now and at every boot, and prints a security notice.

3. **To enable expansion on Wayland**, after reading the section above:

```
sudo usermod -aG input,uinput "$USER"
```

4. **Log out and log back in** (or reboot). Group changes only take effect in a new session.
5. Open **Rheolwyr** from your applications menu.

From source (development)

```
sudo apt install libcairo2-dev libgirepository1.0-dev pkg-config python3-dev xclip wl-clipboard
pip install -r requirements.txt
python -m rheolwyr.main
```

Uninstalling

`sudo apt remove rheolwyr`, then remove yourself from the groups you joined:

```
sudo gpasswd -d "$USER" input
sudo gpasswd -d "$USER" uinput
```

Log out and back in. Delete `~/.local/share/rheolwyr/` and `~/.config/rheolwyr/` to remove your snippets and settings.

5 Quick start

1. Open Rheolwyr. Your snippets are listed on the left; the editor is on the right.
2. Click **+**.
3. Fill in **Snippet Name** (for example `Signature`), **Trigger Text** (for example `;sig`) and **Snippet Content**.
4. Click **Save**. The snippet works immediately.
5. In any text editor, type `;sig`. Rheolwyr erases the trigger and types your text.

> **Tip:** Rheolwyr expands only while it's running, and closing its window quits it. Minimize the window to keep it working in the background.

6 Using Rheolwyr

Writing good triggers

There are two kinds of trigger:

- **Symbol triggers** start with a symbol, such as `;sig` or `+addr`. They expand as soon as you finish typing them, wherever you type them.
- **Word triggers** start with a letter or number, such as `brb` or `addr`. They expand only when you type them as a whole word and press **Space**; Rheolwyr keeps the space. They never fire inside a longer word, so typing "address" doesn't touch `addr`.

Tab doesn't complete a trigger. In forms and browsers Tab moves to another field, where Rheolwyr's backspaces could erase the wrong text, so Tab simply clears what Rheolwyr is matching.

Overlapping symbol triggers. If one symbol trigger is the start of another, such as `;s` and `;sig`, Rheolwyr waits. Type `;sig` and you get the `;sig` snippet. Type `;s` followed by anything else, such as a space, and `;s` expands at that key, which is kept. So `;s` expands one key later than a trigger with no overlap.

How expansion works

1. Rheolwyr erases the trigger with backspaces (plus the Space, for a word trigger).
2. Snippets under 50 characters are **typed** in, using a virtual keyboard on Wayland or X11 input on X11.
3. Longer snippets are **pasted**: Rheolwyr puts the text on the clipboard, presses Ctrl+V, then restores your previous clipboard text. (If your clipboard held an image, it isn't restored.)

Then it re-types any key it erased after the trigger, such as your Space. Rheolwyr ignores its own typing so expansions don't trigger again.

Editing and deleting

Click a snippet to load it, change it, and click **Save**. To delete it, select it and click the trash button in the editor header. Deleting is immediate, so export a backup first if you might want it back.

The menu

Click the menu button (☰):

Item	What it does
System / Light / Dark	Appearance; remembered between runs
Import Snippets	Adds snippets from a JSON file; skips exact duplicates
Export Snippets	Saves all snippets to a JSON file
Instructions	A short how-to
About Rheolwyr	Version, license and credits

7 Privacy and files

Item	Location
Snippets	`~/.local/share/rheolwyr/snippets.db`
Settings	`~/.config/rheolwyr/config.json`
udev rule (from the package)	`/lib/udev/rules.d/99-rheolwyr.rules`
Module loading (from the package)	`/usr/lib/modules-load.d/uinput.conf`

Snippets are stored unencrypted. **Don't keep passwords or other secrets in a text expander.**

8 Limitations

- **Terminals.** Long snippets are pasted with Ctrl+V, which most Linux terminals don't use for paste, so long snippets may not appear there. Short (typed) snippets work.
- **Clipboard history tools** may record long snippets, because they pass through the clipboard.
- **No password-field detection.** Linux has no equivalent of macOS Secure Input, so Rheolwyr can't tell when you're typing a password. It never stores what you type, but quit it first if you'd rather it not see a password at all.
- **Plain text only**, with no formatting, dates or cursor placement.
- **Keyboard layout.** Typed snippets use a US-style key map on Wayland; unusual characters in short snippets may come out differently on other layouts. Long snippets are pasted and aren't affected.
- **Only while running.** Closing the window stops expansion.

9 Troubleshooting

Nothing expands on Wayland. Run `groups` and check for `input` and `uinput`. If they're missing, add them (see *Installing*) and log out and back in. Make sure Rheolwyr is open.

Nothing expands at all. Check `/dev/uinput` exists (`ls -l /dev/uinput`); if not, run `sudo modprobe uinput`.

The trigger disappears but nothing appears. You may be in a terminal with a long snippet; try a text editor.

A word trigger doesn't expand. Press **Space** after it; Tab and Enter don't complete triggers.

A short symbol trigger expands a key late. It's the start of a longer trigger (for example `;s` and `;sig`), so Rheolwyr waits to see which you mean. Rename one of them if that bothers you.

Import says 0 snippets. Every snippet in the file already exists, or the file isn't a Rheolwyr JSON export.

10 Support and license

Report problems on GitHub (github.com/TaliskerMan/Rheolwyr) or email **chuck@nordheim.online**. For security issues, see `SECURITY.md`.

Website	nordheim.online
Source code	github.com/TaliskerMan/Rheolwyr
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